

사이클별 지표 패널

M02Ch103

각 패널 = 한 지표의 cycle 궤적 (routine 0.5C).

early/late 점선 = 앞·뒤 5포인트 median.

트렌드 = 선형 slope/100cyc + aging 기대방향 대조.

주의: LAM_PE / *_proxy / pattern_score는 가설·proxy이며 절대 LAM%/LLI%가 아닙니다. R_ohmic은 $\sqrt{t}$ 외삽 proxy (0.1s 단일값 $\neq$ 순수 음).

트렌드 요약 — M02Ch103

- 유효 지표: 48개 · aging 방향 일치 17 · 반대 1

하락 트렌드 (상위)

- LAM 곡선 proxy: early=-3.11 $\rightarrow$ late=-17.48 (Δ =-14.37, -5.717%/100cyc) $\rightarrow$ decreasing · context
- LLI 곡선 proxy: early=-0.6932 $\rightarrow$ late=-3.752 (Δ =-3.059, -1.527%/100cyc) $\rightarrow$ decreasing · context
- 방전 플래토 폭: early=16.85 $\rightarrow$ late=14.17 (Δ =-2.681, -1.363Q-units/100cyc) $\rightarrow$ decreasing · matches_aging
- R Ω 성장률 /100cyc: early=0.7711 $\rightarrow$ late=0.4781 (Δ =-0.293, -0.2952m Ω /100cyc/100cyc) $\rightarrow$ decreasing · context
- 방전 플래토 ΔV : early=-0.0323 $\rightarrow$ late=-0.07078 (Δ =-0.03847, -0.01613V/100cyc) $\rightarrow$ decreasing · context

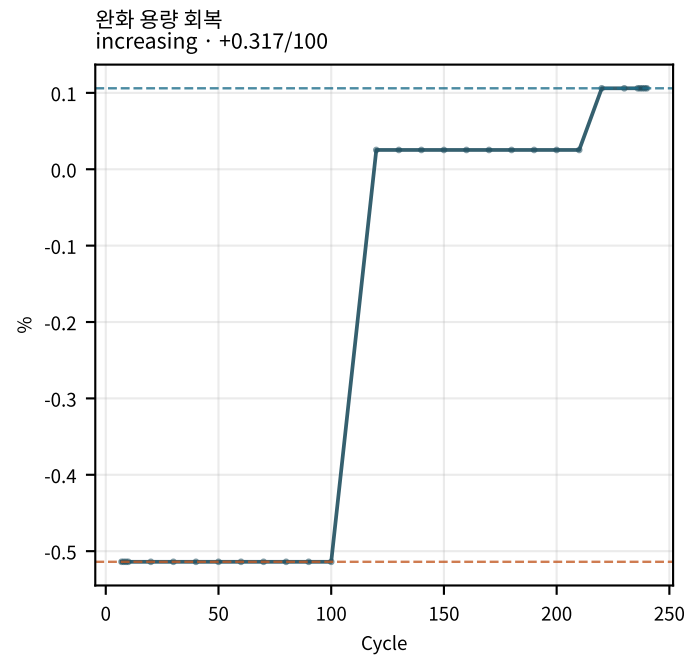
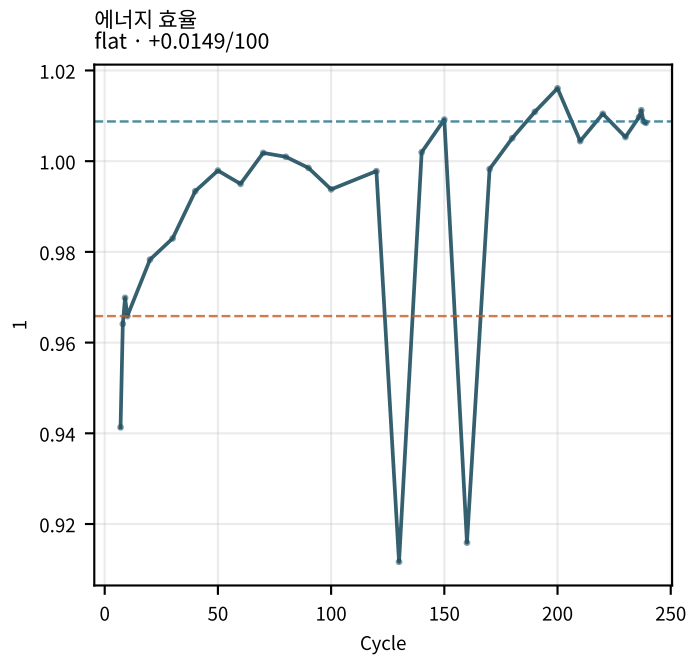
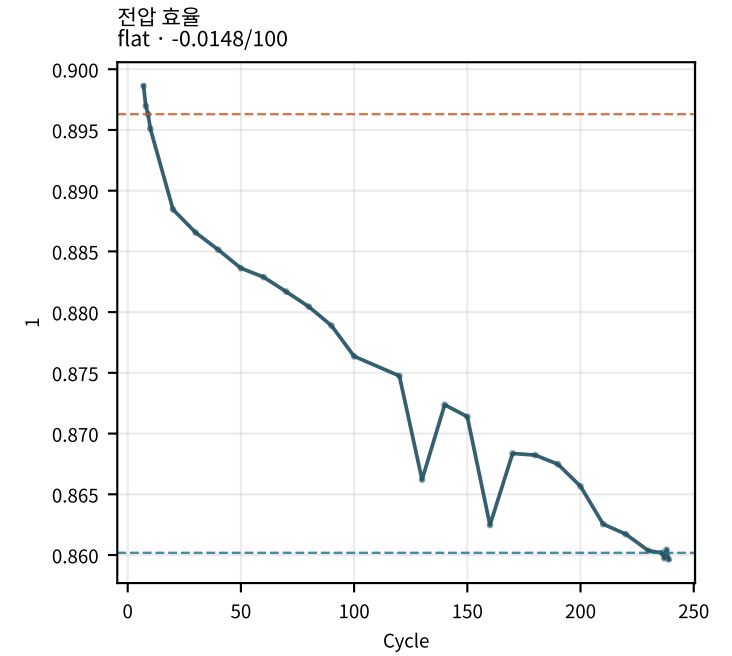
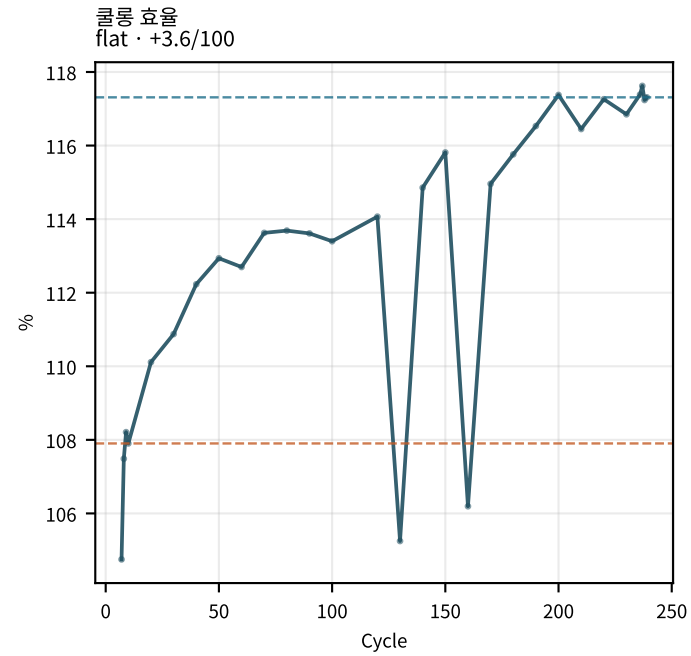
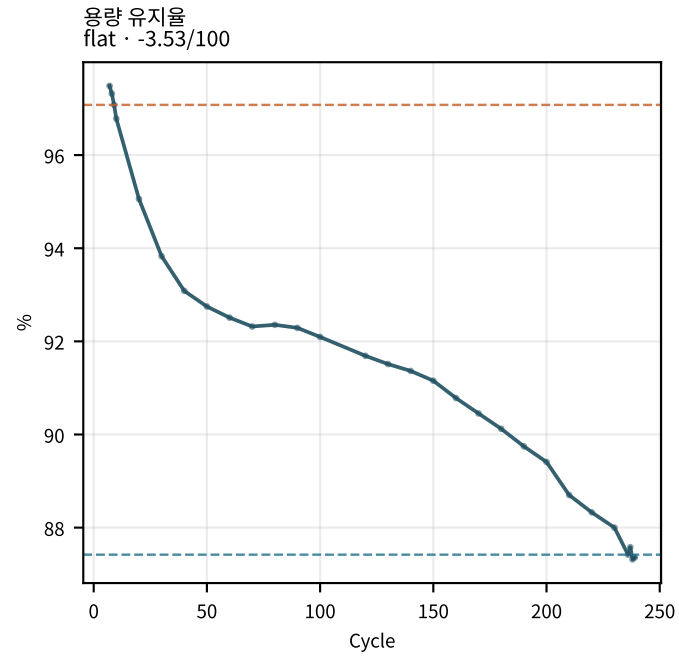
상승 트렌드 (상위)

- 음 저항 (SOC50): early=1.362 $\rightarrow$ late=2.366 (Δ =+1.004, +0.5021m Ω /100cyc) $\rightarrow$ increasing · matches_aging
 - 기계/화학 비: early=1.898 $\rightarrow$ late=2.592 (Δ =+0.6938, +0.37961/100cyc) $\rightarrow$ increasing · matches_aging
 - 완화 용량 회복: early=-0.5139 $\rightarrow$ late=0.106 (Δ =+0.6199, +0.3169%/100cyc) $\rightarrow$ increasing · matches_aging
 - 계면 R 패턴: early=0.453 $\rightarrow$ late=0.8481 (Δ =+0.3951, +0.16420-1/100cyc) $\rightarrow$ increasing · matches_aging
 - LLI 패턴: early=0.374 $\rightarrow$ late=0.6624 (Δ =+0.2883, +0.12770-1/100cyc) $\rightarrow$ increasing · matches_aging
- CycleDiag · plot_cycle_metrics_panel.py

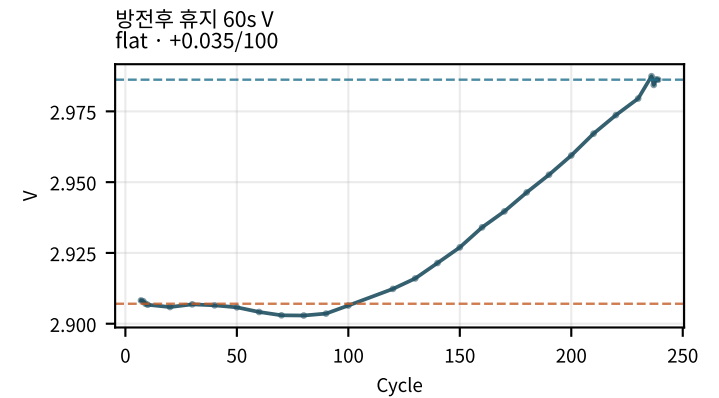
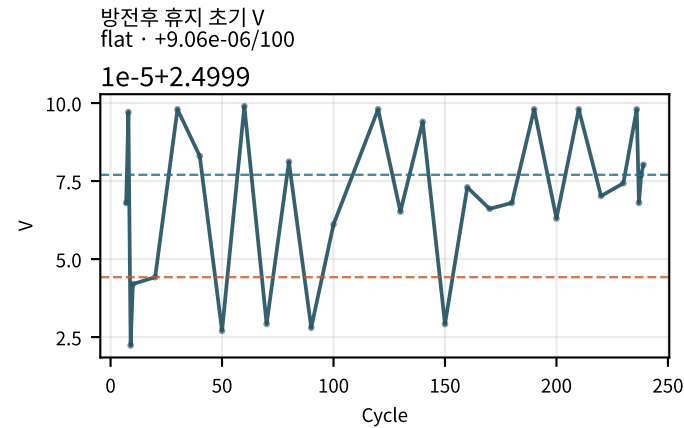
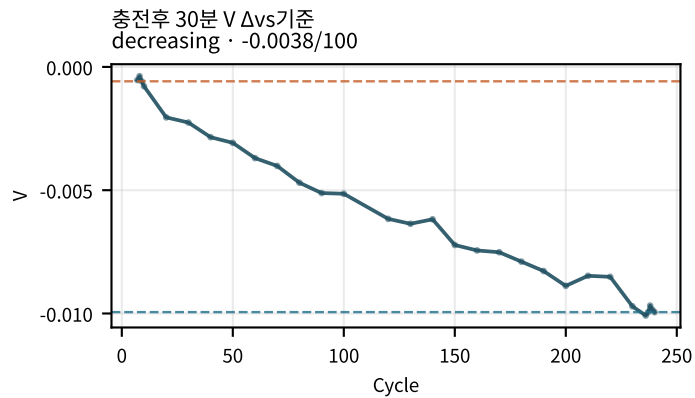
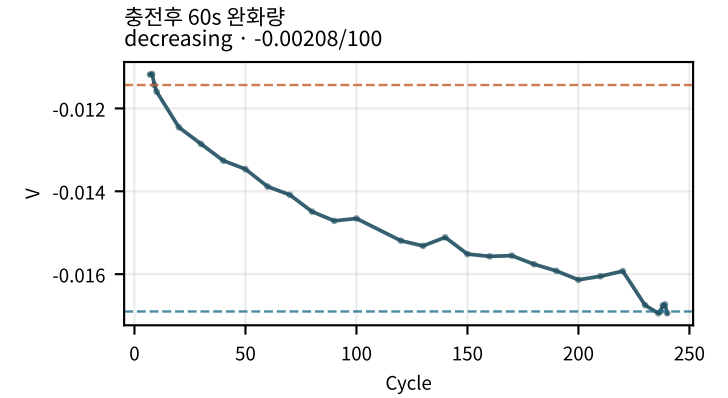
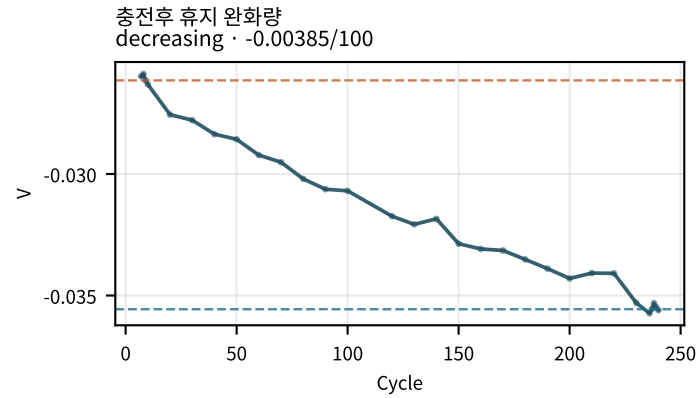
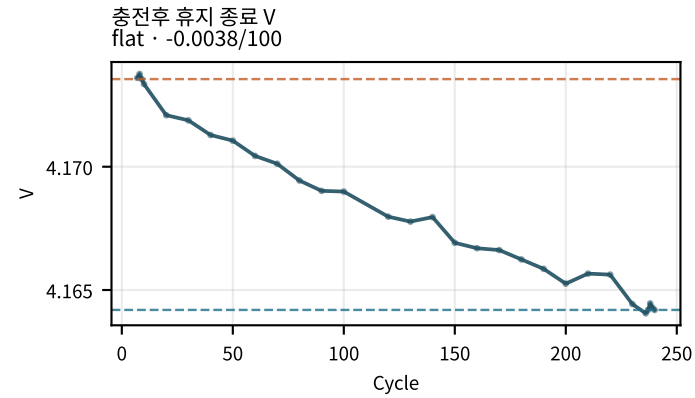
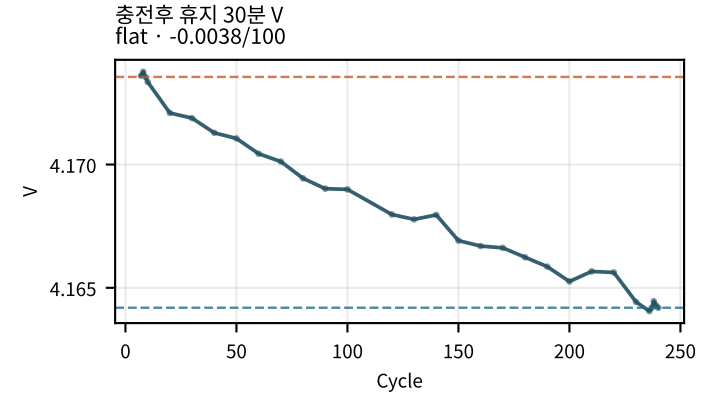
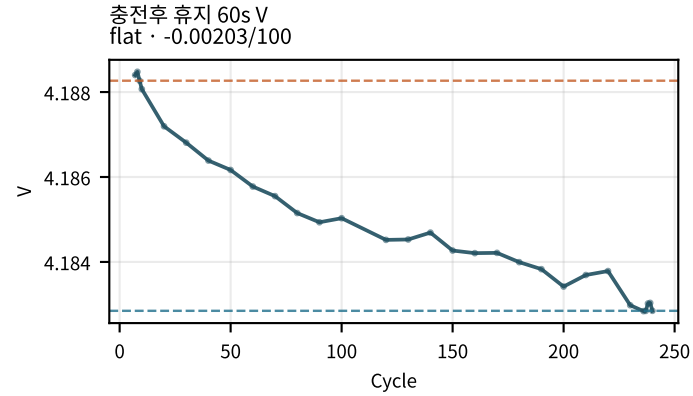
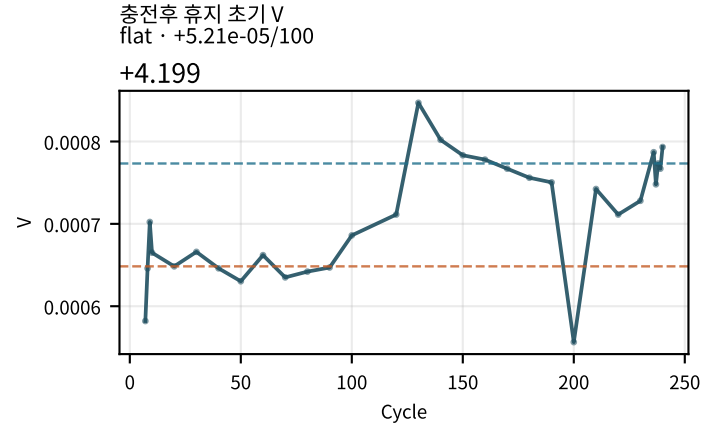
기대 aging과 반대

- 저SOC 히스테리시스: early=0.0766 $\rightarrow$ late=0.0465 (Δ =-0.0301, -0.01222V/100cyc) $\rightarrow$ decreasing · opposite_aging

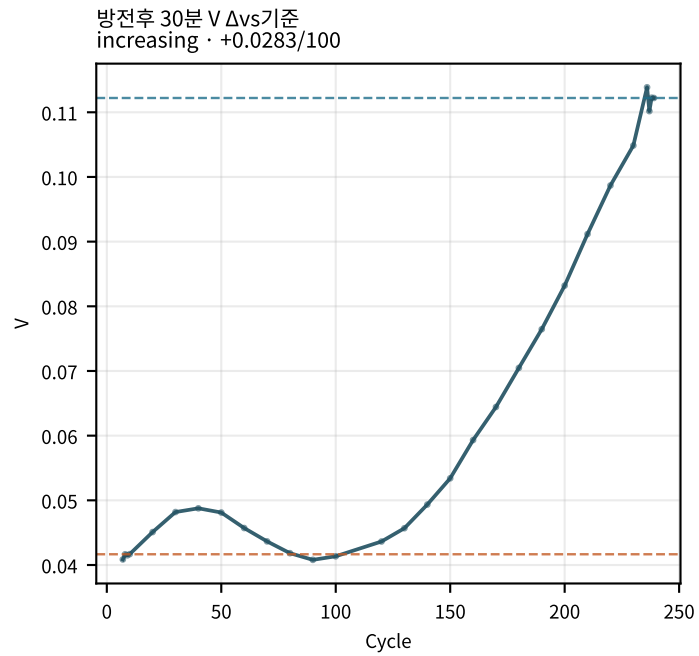
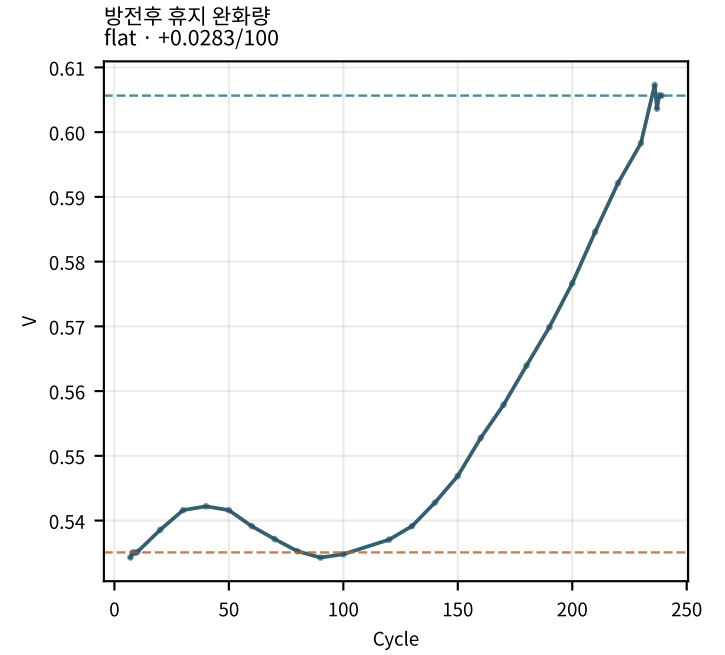
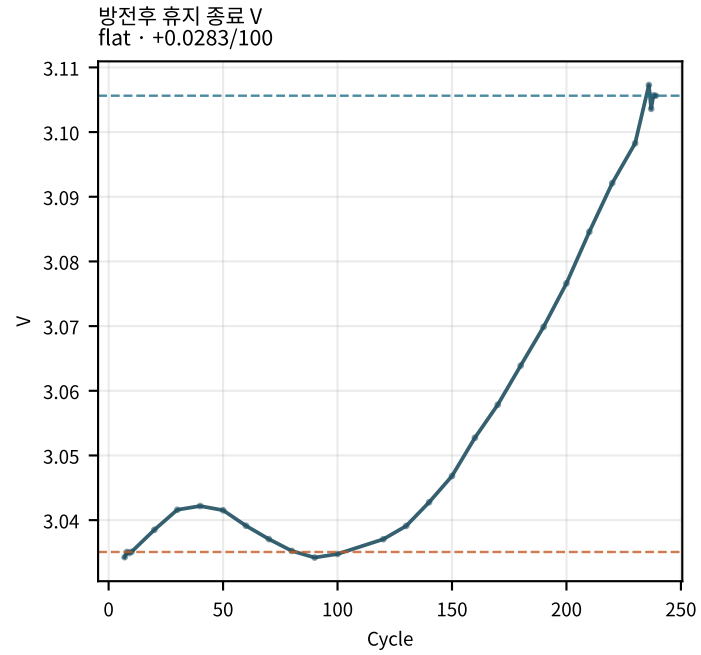
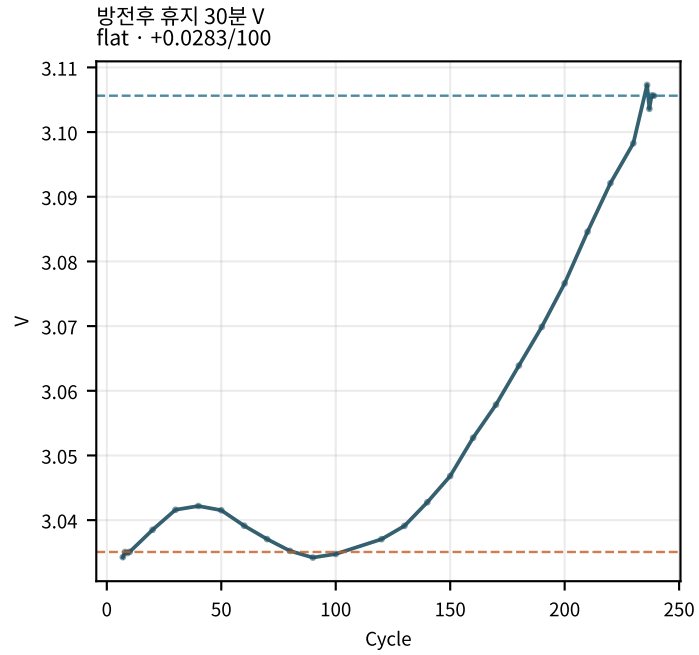
M02Ch103 · 용량 · 효율



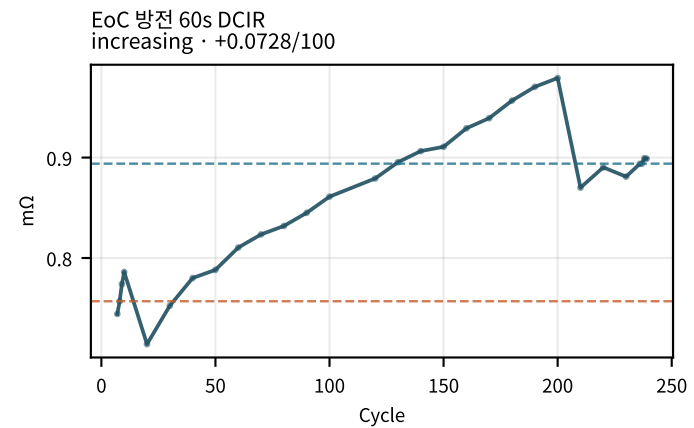
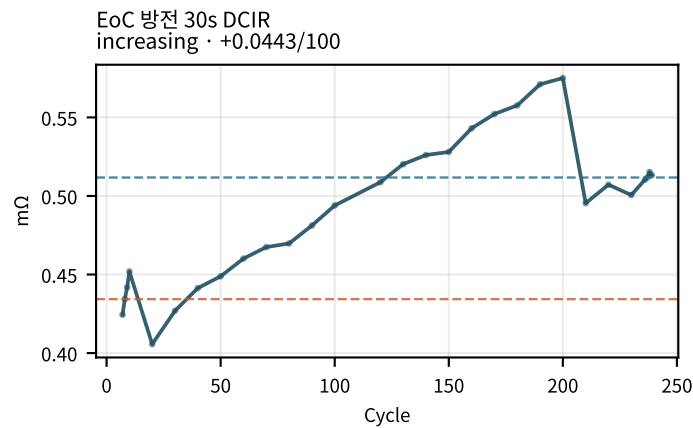
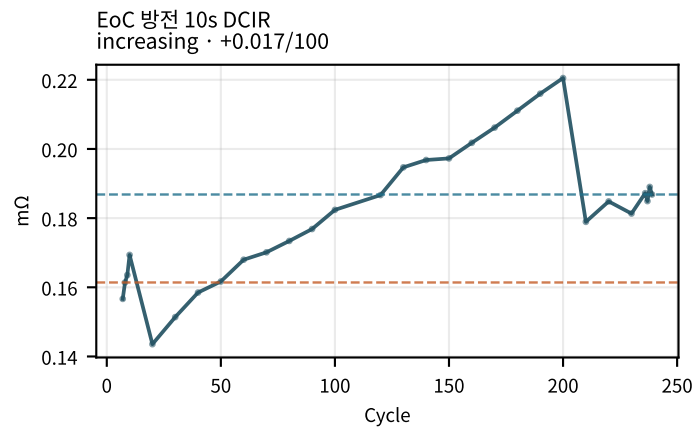
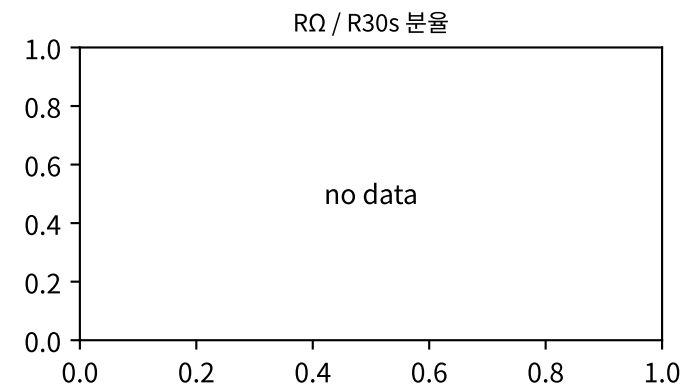
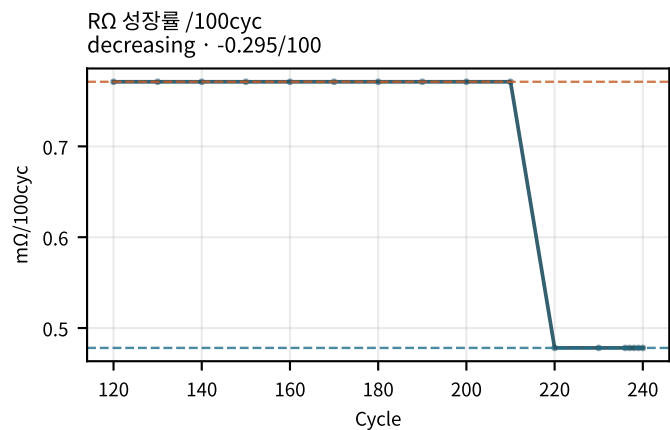
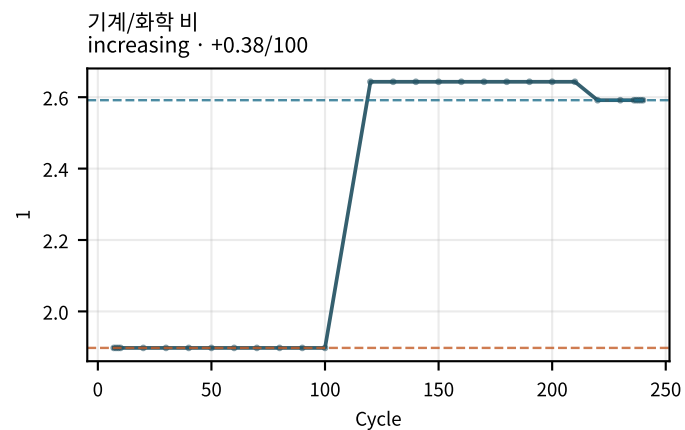
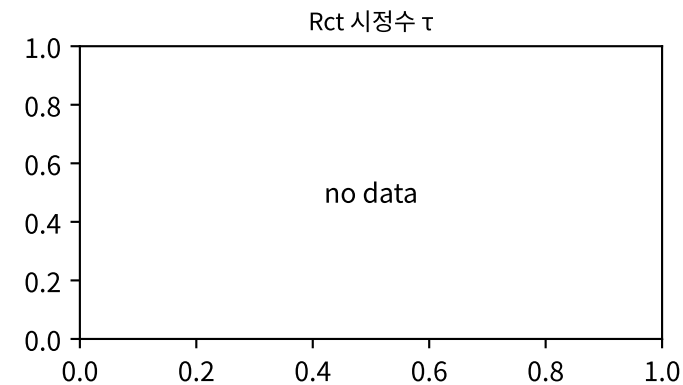
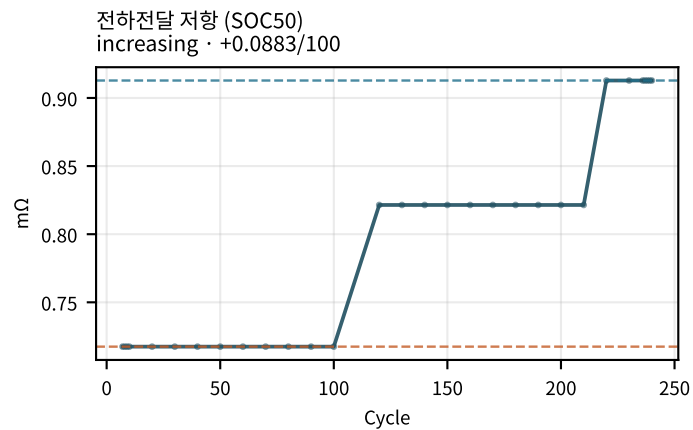
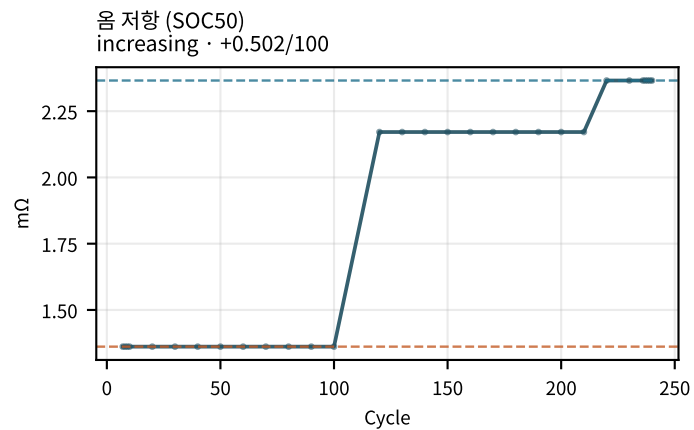
M02Ch103 · 휴지 전압 (충전/방전 후)



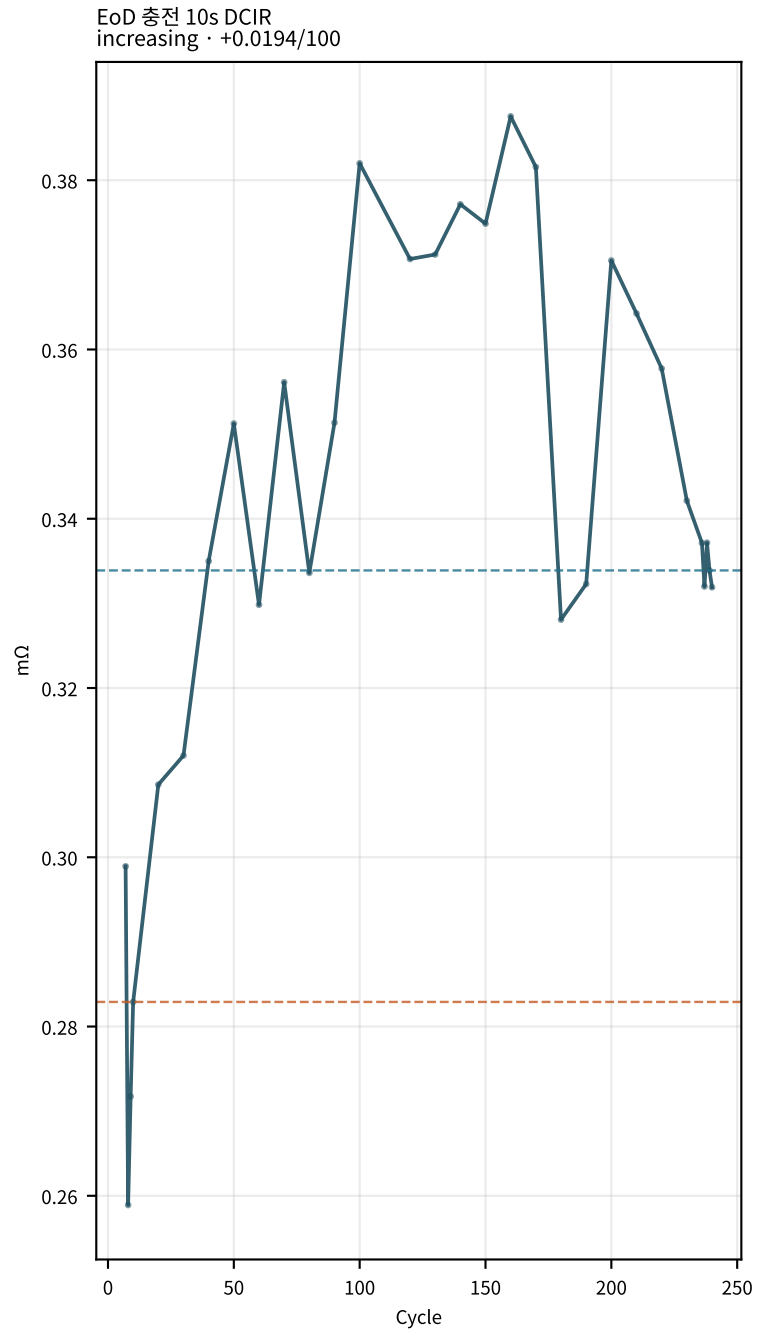
M02Ch103 · 휴지 전압 (충전/방전 후)



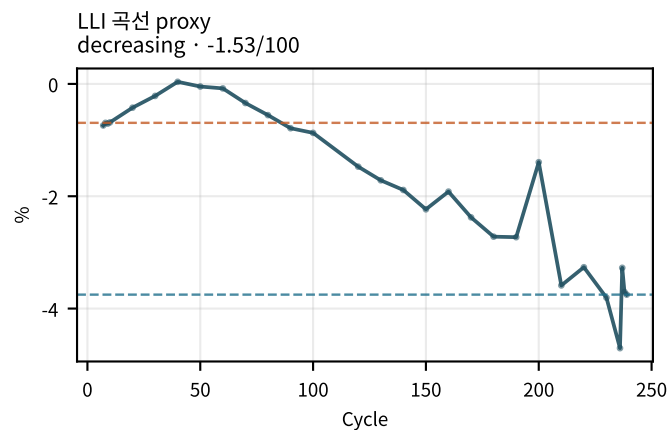
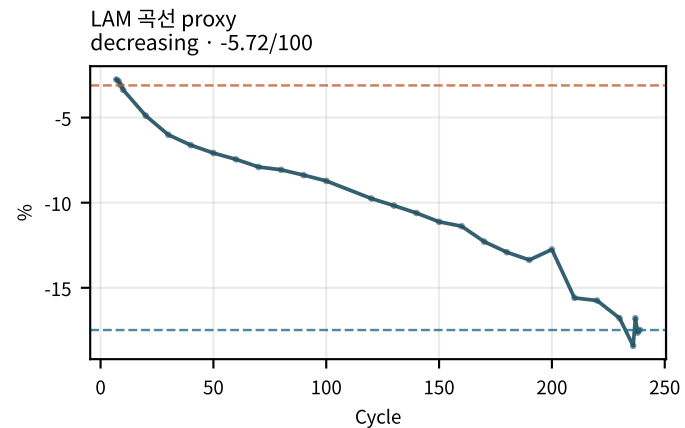
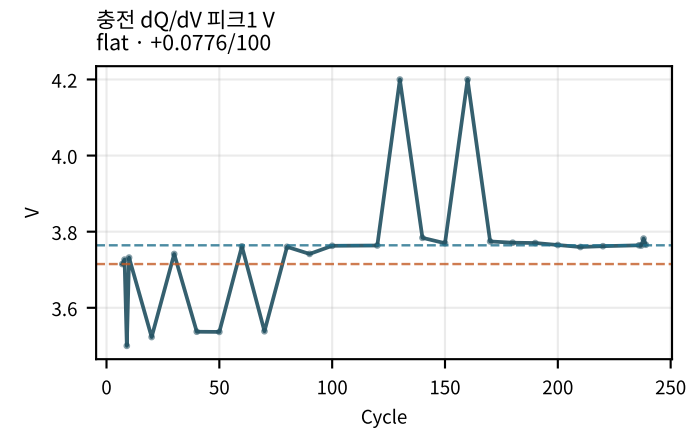
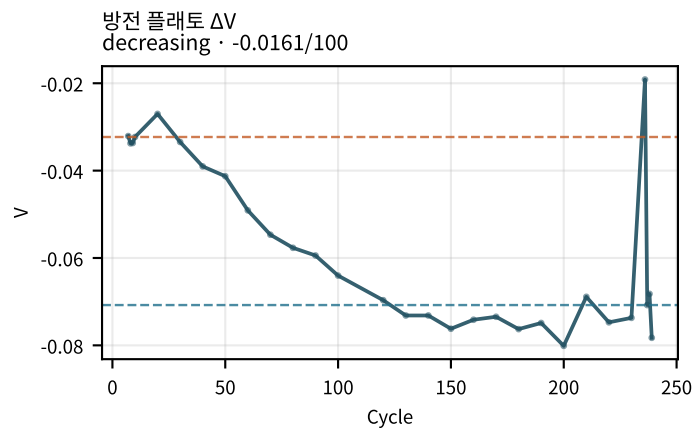
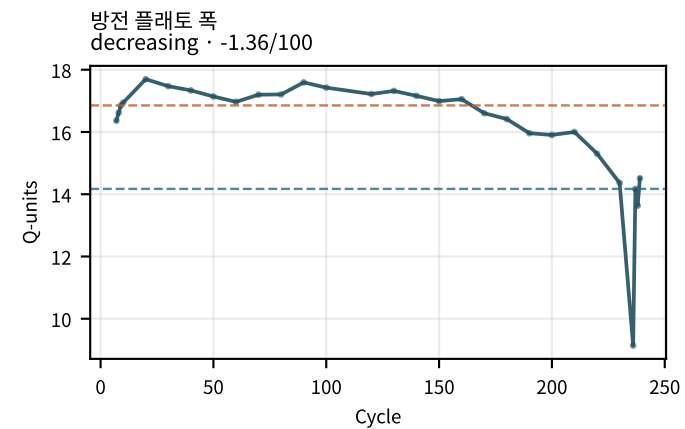
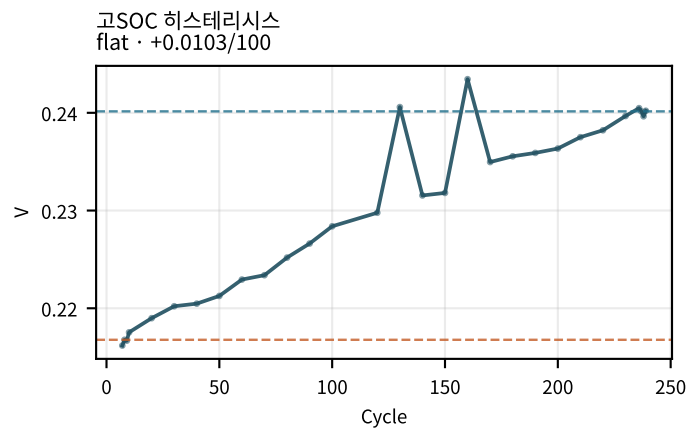
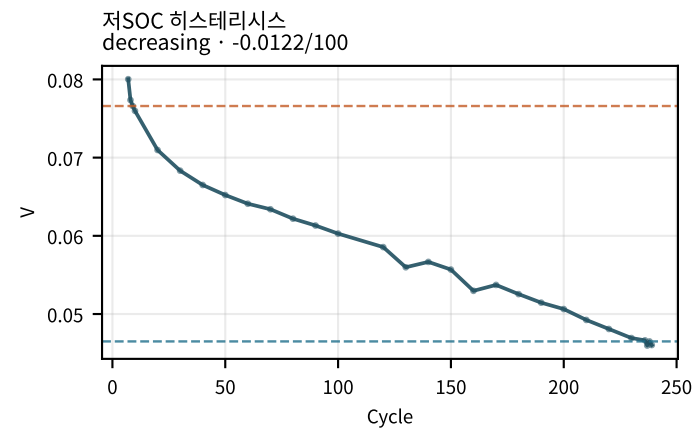
M02Ch103 · 저항 · 분해



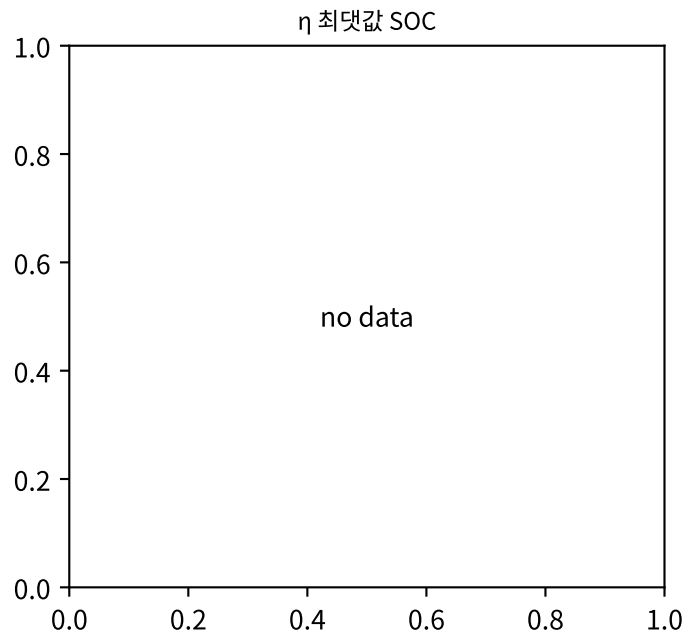
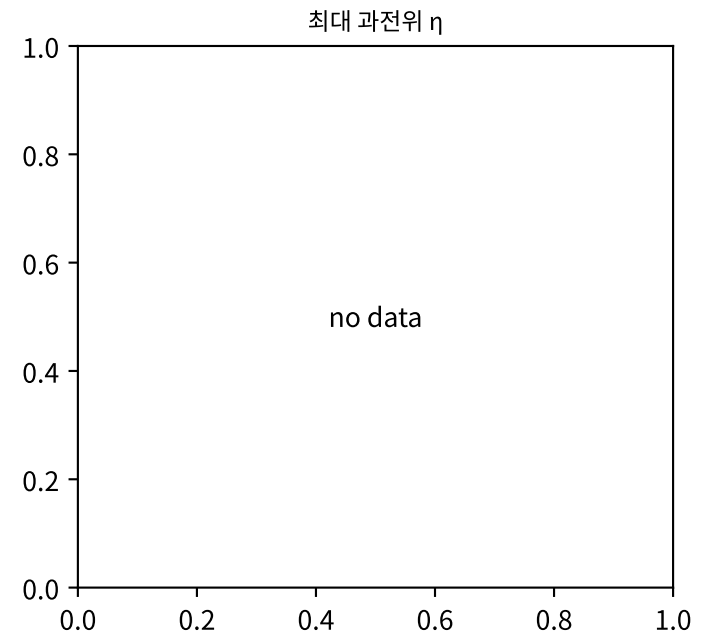
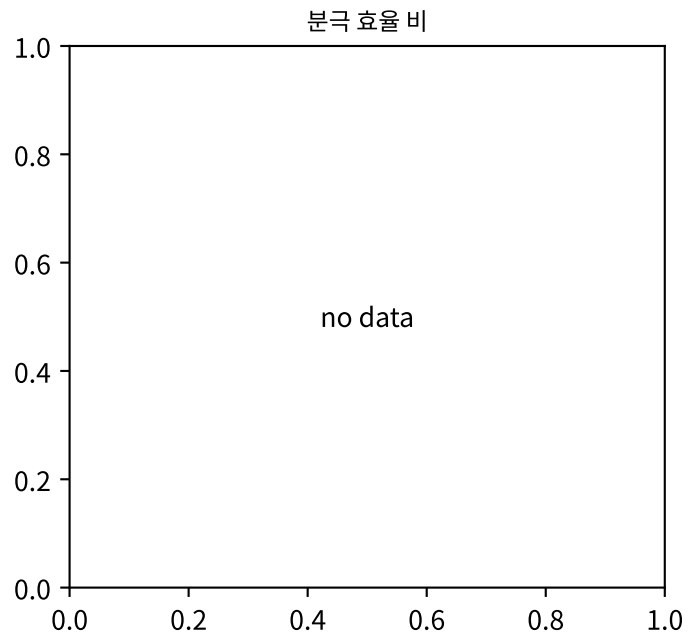
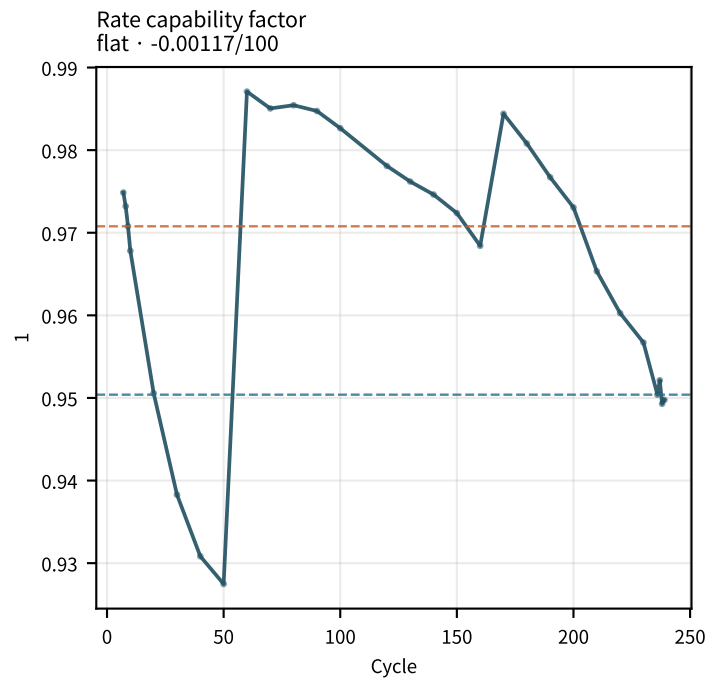
M02Ch103 · 저항 · 분해



M02Ch103 · 곡선 형상 · dQ/dV

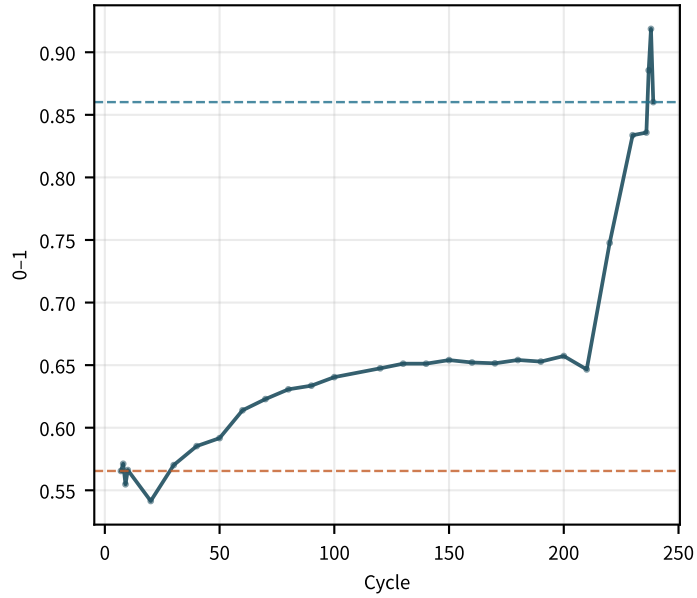


M02Ch103 · 수송 · rate

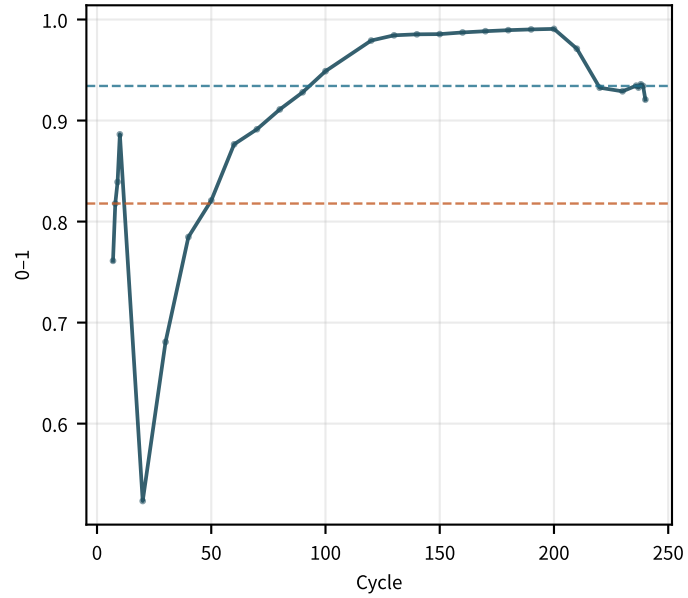


M02Ch103 · 열화 패턴 점수

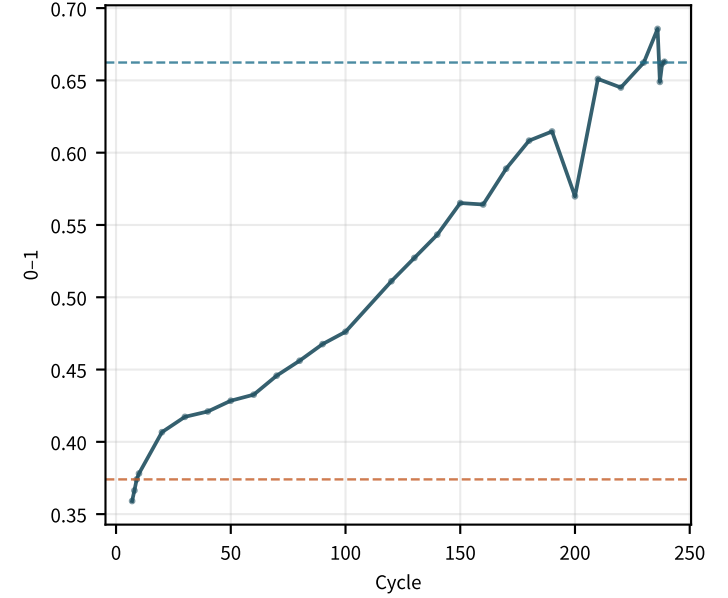
PE activity 패턴
increasing · +0.106/100



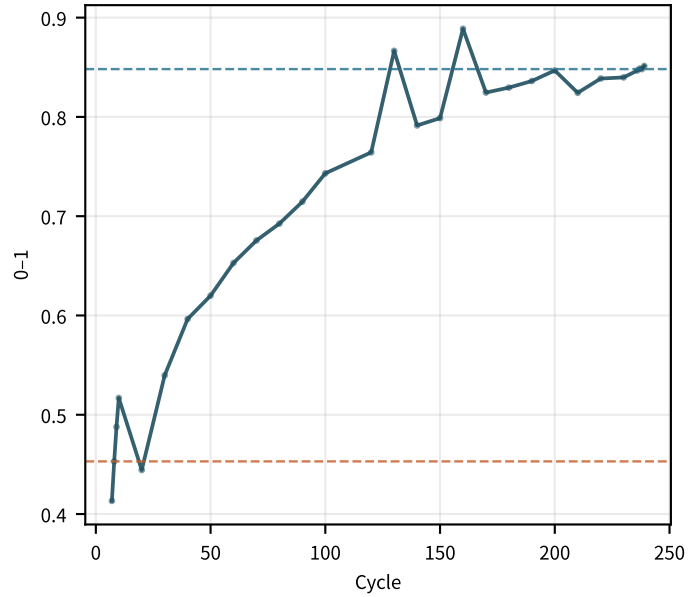
접촉/스택 손실
increasing · +0.0815/100



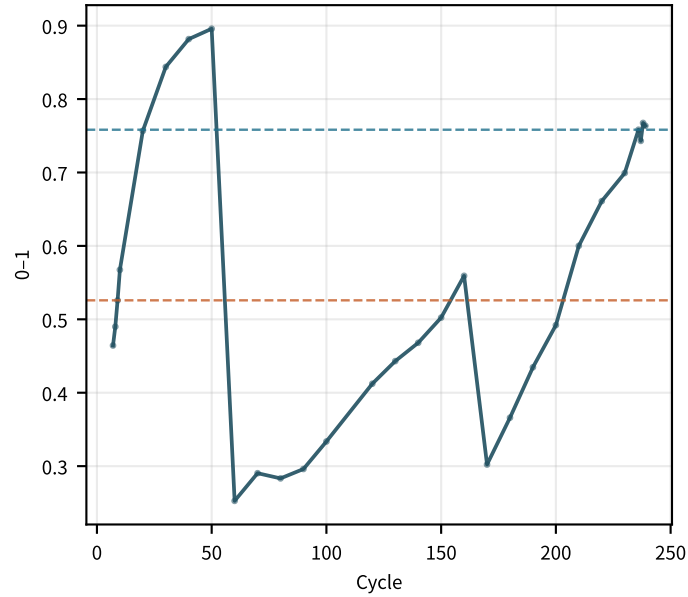
LLI 패턴
increasing · +0.128/100



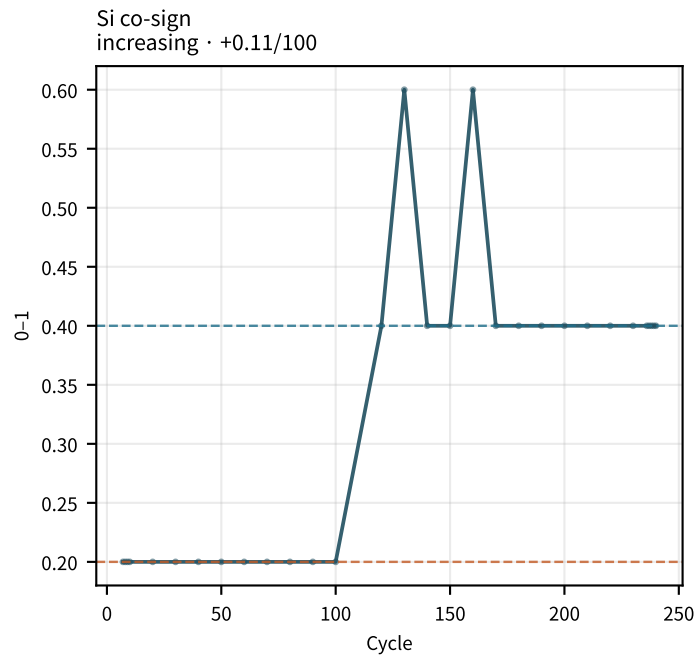
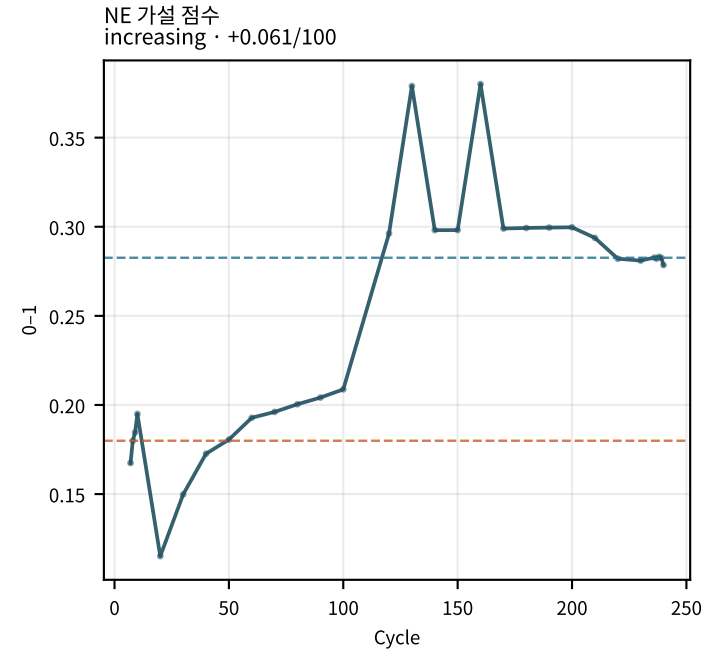
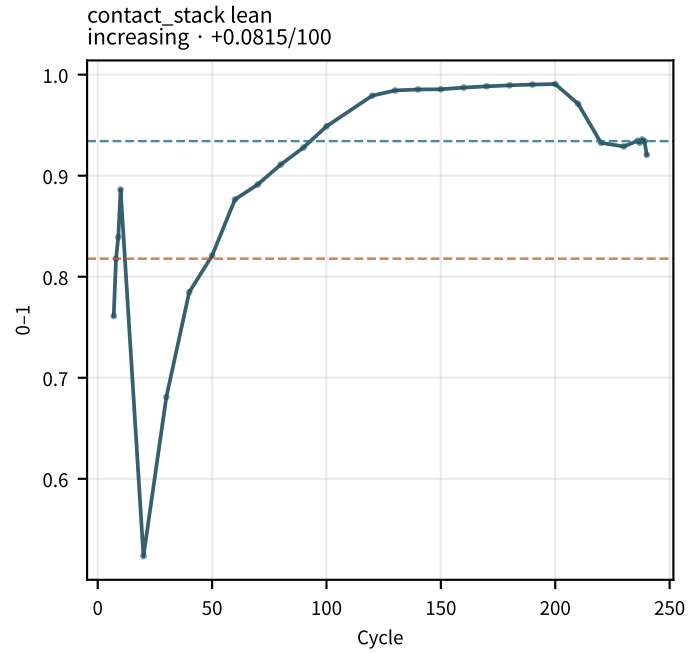
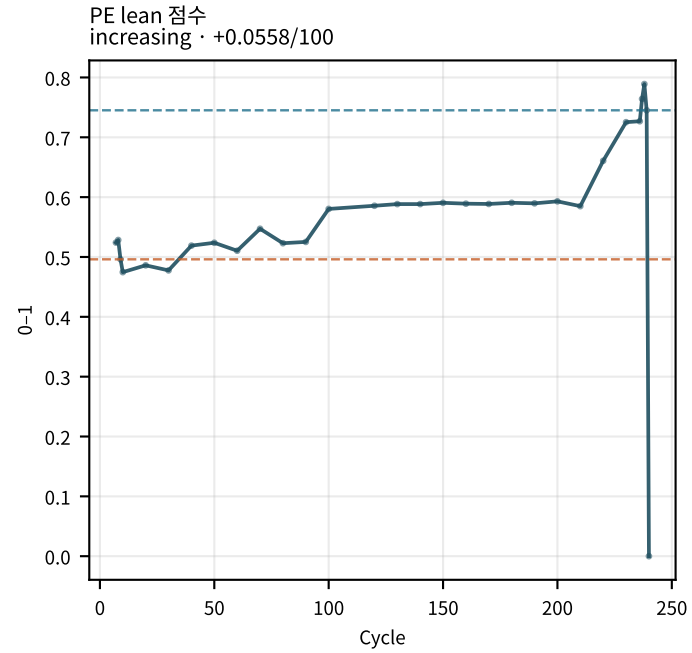
계면 R 패턴
increasing · +0.164/100



고체확산 제한
flat · +0.0327/100



M02Ch103 · 전극 lean 가설



지표 설명 · 트렌드 (M02Ch103)

용량 유지율 (SoHQ)

기준 대비 방전용량 유지율. fade/knee의 1차 관측량. | 계산: $Q_dchg(cycle) / Q_baseline \times 100$ (routine 0.5C).

트렌드: 용량 유지율: early=97.08 → late=87.42 ($\Delta=-9.659$, -3.534%/100cyc) → flat · stable | aging 기대=decrease | vs=stable

쿨롱 효율 (CE)

충전 대비 방전용량 비. LLI·부반응 힌트. | 계산: $Q_dchg / Q_chg \times 100$ (이상치 >102% 또는 <85%는 진단에서 제외).

트렌드: 쿨롱 효율: early=107.9 → late=117.3 ($\Delta=+9.409$, +3.596%/100cyc) → flat · stable | aging 기대=decrease | vs=stable

전압 효율 (VE)

에너지 효율의 전압 측. 저항·분극 증가 시 하락. | 계산: E_dchg / E_chg (Wh).

트렌드: 전압 효율: early=0.8963 → late=0.8602 ($\Delta=-0.03612$, -0.014771/100cyc) → flat · stable | aging 기대=decrease | vs=stable

에너지 효율 (EE)

충·방전 에너지 비. | 계산: E_dchg / E_chg .

트렌드: 에너지 효율: early=0.9658 → late=1.009 ($\Delta=+0.04291$, +0.014881/100cyc) → flat · stable | aging 기대=decrease | vs=stable

완화 용량 회복 (Q_relax_pct)

DCIR/RPT 전후 휴지로 회복되는 용량 분율. Si chemo-mech co-sign. | 계산: DCIR 블록 전후 Q 차이 / Q × 100, routine에 forward-fill.

트렌드: 완화 용량 회복: early=-0.5139 → late=0.106 ($\Delta=+0.6199$, +0.3169%/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

충전후 휴지 초기 V (EoC_restV_init)

충전(EoC) 직후 휴지 시작 전압. | 계산: charge 종료 후 rest step 첫 샘플 전압.

트렌드: 충전후 휴지 초기 V: early=4.2 → late=4.2 ($\Delta=+0.0001249$, +5.205e-05V/100cyc) → flat · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch103)

충전후 휴지 60s V (EoC_restV_60s)

충전 후 휴지 60초 시점 전압. | 계산: rest step_time $\approx$ 60 s 보간.

트렌드: 충전후 휴지 60s V: early=4.188 $\rightarrow$ late=4.183 ($\Delta=-0.005418$, -0.002028V/100cyc) $\rightarrow$ flat · context | aging
기대=either | vs=context

충전후 휴지 30분 V (EoC_restV_30m)

충전 후 휴지 30분 시점 전압. OCV에 가까운 EoC rest. | 계산: rest step_time $\approx$ 1800 s 보간.

트렌드: 충전후 휴지 30분 V: early=4.174 $\rightarrow$ late=4.164 ($\Delta=-0.009361$, -0.003801V/100cyc) $\rightarrow$ flat · context | aging
기대=either | vs=context

충전후 휴지 종료 V (EoC_restV_end)

충전 후 휴지 스텝 마지막 전압. | 계산: rest step 끝 샘플.

트렌드: 충전후 휴지 종료 V: early=4.174 $\rightarrow$ late=4.164 ($\Delta=-0.009361$, -0.003801V/100cyc) $\rightarrow$ flat · context | aging
기대=either | vs=context

충전후 휴지 완화량 (EoC_restV_relax)

휴지 동안 전압 변화 (end – init). 분극 완화. | 계산: EoC_restV_end – EoC_restV_init.

트렌드: 충전후 휴지 완화량: early=-0.02615 $\rightarrow$ late=-0.03556 ($\Delta=-0.009415$, -0.003853V/100cyc) $\rightarrow$ decreasing · context |
aging 기대=either | vs=context

충전후 60s 완화량 (EoC_restV_relax_60s)

휴지 첫 60초 완화량. | 계산: EoC_restV_60s – EoC_restV_init.

트렌드: 충전후 60s 완화량: early=-0.01144 $\rightarrow$ late=-0.0169 ($\Delta=-0.005465$, -0.00208V/100cyc) $\rightarrow$ decreasing · context |
aging 기대=either | vs=context

충전후 30분 V Δ vs기준 (delta_EoC_restV_30m)

기준 사이클 대비 충전후 30분 rest V 이동 (slippage/LLI 힌트). | 계산: EoC_restV_30m(cycle) – EoC_restV_30m(baseline).

트렌드: 충전후 30분 V Δ vs기준: early=-0.0005831 $\rightarrow$ late=-0.009945 ($\Delta=-0.009361$, -0.003801V/100cyc) $\rightarrow$ decreasing ·
context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch103)

방전후 휴지 초기 V (EoD_restV_init)

방전(EoD) 직후 휴지 시작 전압. | 계산: discharge 종료 후 rest step 첫 샘플.

트렌드: 방전후 휴지 초기 V: early=2.5 → late=2.5 (Δ =+3.28e-05, +9.062e-06V/100cyc) → flat · context | aging 기대=either
| vs=context

방전후 휴지 60s V (EoD_restV_60s)

방전 후 휴지 60초 전압. | 계산: rest $\approx$ 60 s 보간.

트렌드: 방전후 휴지 60s V: early=2.907 → late=2.986 (Δ =+0.07914, +0.035V/100cyc) → flat · context | aging 기대=either
| vs=context

방전후 휴지 30분 V (EoD_restV_30m)

방전 후 휴지 30분 전압. OCV에 가까운 EoD rest. | 계산: rest $\approx$ 1800 s 보간.

트렌드: 방전후 휴지 30분 V: early=3.035 → late=3.106 (Δ =+0.07055, +0.02831V/100cyc) → flat · context | aging
기대=either | vs=context

방전후 휴지 종료 V (EoD_restV_end)

방전 후 휴지 스텝 마지막 전압. | 계산: rest step 끝 샘플.

트렌드: 방전후 휴지 종료 V: early=3.035 → late=3.106 (Δ =+0.07055, +0.02831V/100cyc) → flat · context | aging 기대=either
| vs=context

방전후 휴지 완화량 (EoD_restV_relax)

방전 후 휴지 완화 (end – init). | 계산: EoD_restV_end – EoD_restV_init.

트렌드: 방전후 휴지 완화량: early=0.5351 → late=0.6056 (Δ =+0.07057, +0.0283V/100cyc) → flat · context | aging 기대=either
| vs=context

방전후 30분 V Δ vs기준 (delta_EoD_restV_30m)

기준 대비 방전후 30분 rest V 이동. | 계산: EoD_restV_30m(cycle) – baseline.

트렌드: 방전후 30분 V Δ vs기준: early=0.04166 → late=0.1122 (Δ =+0.07055, +0.02831V/100cyc) → increasing · context |
aging 기대=either | vs=context

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옴 저항 (SOC50) (R_ohmic_soc50)

DCIR 펄스 early $\sqrt{t}$ 외삽 $t \rightarrow 0$ 절편. 순수 옴이 아니라 초기 비옴 성분 proxy. | 계산: $R(t) = |\Delta V|/|I|$; $t \in [0, 0.3]$ s 에서 R vs $\sqrt{t}$ 회귀 절편.
SOC50 블록 $\rightarrow$ forward-fill.

트렌드: 옴 저항 (SOC50): early=1.362 $\rightarrow$ late=2.366 (Δ =+1.004, +0.5021m Ω /100cyc) $\rightarrow$ increasing · matches_aging |
aging 기대=increase | vs=matches_aging

전하전달 저항 (SOC50) (R_ct_soc50)

중간 시간대 잔차의 지수 포화항. 계면/화학 저항 proxy (Cdl 직접 추정 아님). | 계산: $\text{resid} = R - R_{\Omega} - A\sqrt{t}$; $t \in [0.3, 10]$ s 에서 $R_{ct}(1 - e^{-(t/\tau)})$ fit.

트렌드: 전하전달 저항 (SOC50): early=0.7175 $\rightarrow$ late=0.9128 (Δ =+0.1953, +0.08828m Ω /100cyc) $\rightarrow$ increasing · matches_aging |
aging 기대=increase | vs=matches_aging

Rct 시정수 τ (tau_ct_soc50)

$R(t)$ 잔차 fit의 유효 시정수. $\tau \approx R_{ct} \cdot C_{dl}$ 가설이지만 Cdl은 분리하지 않음. | 계산: curve_fit 초기값 2s, bounds [0.05, 20]; 데이터로 추정.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

기계/화학 비 (mech_vs_chem_ratio)

옴(스택·접촉) vs 전하전달(계면) 상대 비중. | 계산: $R_{\text{ohmic_soc50}} / R_{\text{ct_soc50}}$.

트렌드: 기계/화학 비: early=1.898 $\rightarrow$ late=2.592 (Δ =+0.6938, +0.37961/100cyc) $\rightarrow$ increasing · matches_aging | aging
기대=increase | vs=matches_aging

R Ω 성장률 /100cyc (R_ohmic_growth_100)

기준 DCIR 대비 100사이클당 옴 저항 증가율. 레벨(R_{Ω})과 별개 — 감속해도 R_{Ω} 는 계속 상승 가능. | 계산: $(R_{\Omega} - R_{\Omega 0}) / ((\text{cyc} - \text{cyc0})/100)$.

트렌드: R Ω 성장률 /100cyc: early=0.7711 $\rightarrow$ late=0.4781 (Δ =-0.293, -0.2952m Ω /100cyc/100cyc) $\rightarrow$ decreasing · context |
aging 기대=either | vs=context

R Ω / R30s 분율 (R_ohmic_frac_soc50)

30s 총저항 중 옴 성분 비중. | 계산: $R_{\text{ohmic}} / R_{\text{30s_total}}$ @ SOC50.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

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EOC 방전 10s DCIR (EoC_dchgR_10s)

충전 종료 후 방전 시작 10초 시점 $\Delta V/I$. Rct·확산 포함 총 DCIR. | 계산: $|V_0 - V(10s)|/|I| \cdot 1000$. 순수 R Ω 아님.

트렌드: EoC 방전 10s DCIR: early=0.1614 → late=0.1868 (Δ =+0.02544, +0.01702m Ω /100cyc) → increasing · matches_aging
| aging 기대=increase | vs=matches_aging

EOC 방전 30s DCIR (EoC_dchgR_30s)

충전 후 방전 30초 시점 DCIR. | 계산: $|V_0 - V(30s)|/|I| \cdot 1000$.

트렌드: EoC 방전 30s DCIR: early=0.4343 → late=0.5117 (Δ =+0.0774, +0.04427m Ω /100cyc) → increasing · matches_aging
| aging 기대=increase | vs=matches_aging

EOC 방전 60s DCIR (EoC_dchgR_60s)

충전 후 방전 60초 시점 DCIR. | 계산: $|V_0 - V(60s)|/|I| \cdot 1000$.

트렌드: EoC 방전 60s DCIR: early=0.757 → late=0.8939 (Δ =+0.137, +0.0728m Ω /100cyc) → increasing · matches_aging
| aging 기대=increase | vs=matches_aging

EOd 충전 10s DCIR (EoD_chgR_10s)

방전 종료 후 충전 시작 10초 DCIR. | 계산: $|V_0 - V(10s)|/|I| \cdot 1000$.

트렌드: EoD 충전 10s DCIR: early=0.2829 → late=0.3339 (Δ =+0.05099, +0.01935m Ω /100cyc) → increasing · matches_aging
| aging 기대=increase | vs=matches_aging

저SOC 히스테리시스 (hyst_area_low)

저SOC 충·방전 전압 면적. Si-on-Gr chemo-mech 지표. | 계산: V-Q 히스테리시스를 SOC 밴드별로 적분 (low band).

트렌드: 저SOC 히스테리시스: early=0.0766 → late=0.0465 (Δ =-0.0301, -0.01222V/100cyc) → decreasing · opposite_aging
| aging 기대=increase | vs=opposite_aging

고SOC 히스테리시스 (hyst_area_high)

고SOC 히스테리시스. PE activity 보조 증거. | 계산: 히스테리시스 high-SOC band 면적.

트렌드: 고SOC 히스테리시스: early=0.2168 → late=0.2401 (Δ =+0.02337, +0.01028V/100cyc) → flat · stable | aging
기대=increase | vs=stable

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방전 플래토 폭 (dchg_plateau_width)

방전 플래토 구간 폭. 좁아지면 PE isolation/activity 패턴. | 계산: 방전 V 플래토 검출 후 Q 폭.

트렌드: 방전 플래토 폭: early=16.85 → late=14.17 ($\Delta=-2.681$, $-1.363Q\text{-units}/100\text{cyc}$) → decreasing · matches_aging | aging 기대=decrease | vs=matches_aging

방전 플래토 ΔV (delta_dchg_plateau_V)

기준 대비 방전 플래토 전압 이동. | 계산: plateau_V(cycle) - plateau_V(baseline).

트렌드: 방전 플래토 ΔV : early=-0.0323 → late=-0.07078 ($\Delta=-0.03847$, $-0.01613V/100\text{cyc}$) → decreasing · context | aging 기대=either | vs=context

충전 dQ/dV 피크1 V (chg_dQdV_peak1_V)

충전 IC 1번 피크 위치. FC-OCP 매칭·위상 이동 추적. | 계산: Q 보간 → SG → find_peaks.

트렌드: 충전 dQ/dV 피크1 V: early=3.715 → late=3.764 ($\Delta=+0.04926$, $+0.07757V/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

LAM 곡선 proxy (LAM_curve_proxy)

기준 V-Q 대비 scale 축소 proxy. 절대 LAM% 아님; 부호·bound 포화에 주의. | 계산: $(1 - \text{fit_scale}) * 100$ (discharge 3-param fit). scale>1 이면 음수.

트렌드: LAM 곡선 proxy: early=-3.11 → late=-17.48 ($\Delta=-14.37$, $-5.717\%/100\text{cyc}$) → decreasing · context | aging 기대=either | vs=context

LLI 곡선 proxy (LLI_curve_proxy)

기준 대비 Q축 offset. 절대 LLI% 아님. | 계산: $\text{fit_offset} / Q_{\text{max}} \times 100$.

트렌드: LLI 곡선 proxy: early=-0.6932 → late=-3.752 ($\Delta=-3.059$, $-1.527\%/100\text{cyc}$) → decreasing · context | aging 기대=either | vs=context

Rate capability factor (RCF)

0.5C 용량 / 가까운 C/3 RPT 용량. rate 손실 시 하락. | 계산: $Q_{0.5C(N)} / Q_{C/3}(\text{nearest RPT})$.

트렌드: Rate capability factor: early=0.9708 → late=0.9504 ($\Delta=-0.02038$, $-0.0011661/100\text{cyc}$) → flat · stable | aging 기대=decrease | vs=stable

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분극 효율 비 (PER)

과전위 대비 DCIR 스케일. 수송 제한 시 변화. | 계산: $\eta(\text{SOC50}) / (\Delta I \cdot R_{\text{DCIR}_50})$.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

최대 과전위 η (eta_max)

C/3 vs 0.5C 동일 Q속 최대 전압차. | 계산: 동일 Q에서 $|V_{\text{C3}} - V_{\text{0.5C}}|$ 최댓값.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

η 최댓값 SOC (eta_argmax_SOC)

η 가 최대인 SOC. 고SOC면 PE, 저SOC면 NE 쪽 힌트. | 계산: argmax_SOC of $\eta(\text{SOC})$.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

PE activity 패턴 (LAM_PE_pattern_score)

NCM 이차 고립/activity 패턴. 절대 LAM% 금지. | 계산: plateau·dQdV·LAM_curve_proxy·dQV_log_var 가중합 (mode_weights).

트렌드: PE activity 패턴: early=0.5654 → late=0.8602 ($\Delta=+0.2947$, $+0.10610-1/100\text{cyc}$) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

접촉/스택 손실 (contact_loss_score)

옴 성장·분율·mech/chem 증가 증거 합. 전극 미분해. | 계산: R_ohmic_growth, $\Delta R\Omega$, $R\Omega_{\text{frac}}$, EoC_10s, mech/chem 가중합.

트렌드: 접촉/스택 손실: early=0.8179 → late=0.9342 ($\Delta=+0.1164$, $+0.08150-1/100\text{cyc}$) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

LLI 패턴 (LLI_pattern_score)

CE·slippage·곡선 offset 기반 LLI 가설 점수. | 계산: CE ↓, CI ↑, restV/cutoff margin, LLI_curve_proxy 등.

트렌드: LLI 패턴: early=0.374 → late=0.6624 ($\Delta=+0.2883$, $+0.12770-1/100\text{cyc}$) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

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계면 R 패턴 (interface_R_score)

Rct· τ ·VE 하락 등 계면저항 성장 가설. | 계산: R_ct ↑, tau_ct ↑, VE ↓, EoC_60s 등.

트렌드: 계면 R 패턴: early=0.453 → late=0.8481 (Δ =+0.3951, +0.16420–1/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

고체확산 제한 (solid_diffusion_score)

A_diff·PER·RCF 기반 확산/수송 제한. | 계산: A_diff ↑, PER ↑, RCF ↓ 등.

트렌드: 고체확산 제한: early=0.5259 → late=0.7582 (Δ =+0.2324, +0.032690–1/100cyc) → flat · stable | aging 기대=increase | vs=stable

PE lean 점수 (PE_side_score)

LAM_PE_pattern + feature boost + FC-OCP Δ hits. | 계산: 0.75 · LAM_PE + 0.20 · PE_boost + peak_boost.

트렌드: PE lean 점수: early=0.4961 → late=0.7451 (Δ =+0.249, +0.055840–1/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

contact_stack lean (contact_stack_score)

contact_loss를 dominant 경쟁에 넣는 이름. $\approx$ contact_loss_score. | 계산: clip(contact_loss_score, 0, 1).

트렌드: contact_stack lean: early=0.8179 → late=0.9342 (Δ =+0.1164, +0.08150–1/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

NE 가설 점수 (NE_side_score)

contact × Si chemo-mech co-sign. Si 없으면 NE 확정 금지. | 계산: 0.55 · contact · (0.25+0.75 · si) + 0.30 · NE_boost · si.

트렌드: NE 가설 점수: early=0.1799 → late=0.2826 (Δ =+0.1027, +0.061030–1/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

Si co-sign (si_cosign)

저SOC hyst·Q_relax·mech/chem·CV 등 Si chemo-mech 동시 신호. | 계산: SI_NE_COSIGN feature boost (baseline 대비 ↑ 비율).

트렌드: Si co-sign: early=0.2 → late=0.4 (Δ =+0.2, +0.110–1/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging